

GL Towers Dashboard

Operational Threshold Specification

****Version 1.0****

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1. Purpose

This document is the official engineering reference for all operational thresholds used in the GL Towers Dashboard Version 1.0.

All threshold values in this document are extracted directly from the implemented dashboard source. No values have been estimated, approximated, or invented.

Where thresholds intentionally do not yet exist in the current implementation, this is explicitly stated as "Reserved for future implementation."

This document is authoritative for firmware implementation. Any discrepancy between this document and the dashboard source code constitutes either a documentation error or a source code error — both must be resolved before firmware is released.

2. How to Read This Document

Each section covers one monitored parameter. Sections include:

- **Master threshold table** — the complete range classification with all boundary values
- **Alert priority table** — which conditions generate which alert severity
- **Dashboard display table** — how each state is represented on screen
- **Recommended action** — the prescribed operator response per condition
- **Notes** — engineering context, implementation details, or GL Towers observations

Boundary conditions: ranges are specified as mathematically closed or open intervals. The notation "< X" means strictly less than X. "≤ X" means less than or equal to X. Unless otherwise stated, the lower bound is inclusive.

3. pH

3.1 Sensor

Attribute	Value
Parameter	pH of nutrient solution
Unit	pH units (dimensionless)
Sensor	pH probe in recirculating nutrient reservoir
Telemetry field	`sensors.ph`
Display precision	2 decimal places

3.2 Threshold Table

Range	Label	Status Code	Dashboard Color	Dashboard State	Alert Priority
< 5.5	Critical low	warn	Red	Critical	Critical
5.5 – < 5.8	Slightly low	low	Amber (subtle)	Monitoring	Informational
5.8 – ≤ 6.5	Optimal	ok	Green	Stable	None
> 6.5 – ≤ 7.2	Slightly high	low	Amber (subtle)	Monitoring	Informational
> 7.2 – ≤ 7.5	High	warn	Amber	Attention	Warning
> 7.5	Critical high	warn	Red	Critical	Critical

3.3 Agronomist Rule Mapping

Rule ID	Condition	Severity	Priority	Recommended Action
ph.too_low	ph < 5.5	critical	20	Raise pH immediately using pH-Up. Recheck every 15 min until stable within 5.8–6.5.

ph.slightly_low	$5.5 \leq \text{ph} < 5.8$	info	22	Monitor pH trend. If continues falling, add small pH-Up, recheck after 30 min.
ph.slightly_high	$6.5 < \text{ph} \leq 7.2$	info	24	Continue monitoring. If above 6.5 for extended periods, consider small pH-Down.
ph.high	$7.2 < \text{ph} \leq 7.5$	warning	23	Add pH-Down in small increments. Recheck after 30 min before adding more.
ph.too_high	$\text{ph} > 7.5$	critical	21	Reduce pH immediately using pH-Down. Recheck every 15 min until stable within 5.8–6.5.

3.4 Alert Generation

Condition	Alert dedup key	Alert Severity
-----	-----	-----
pH insight = 'warn'	ph-warn-{deviceId}	warning
pH insight = 'low'	ph-low-{deviceId}	warning

3.5 Notes

- The range 6.5–7.2 is classified as "Slightly high" (informational, not warning). GL Towers field testing demonstrated continued operational stability up to approximately pH 7.2. This range is intentionally less strict than standard hydroponic references.
- The range 7.2–7.5 is the "Attention" zone where nutrient lockout begins to occur.
- Above pH 7.5, major nutrients are unavailable to plants and immediate correction is required.
- The `sensors.ph.warnHi = 7.0` value in CONFIG.sensors is used for display-level alerts. The agronomist rule engine uses the more detailed `CONFIG.agronomist.ph` values as listed above.

4. Water Temperature

4.1 Sensor

Attribute	Value

Parameter	Temperature of nutrient solution
Unit	Degrees Celsius (°C)
Sensor	Temperature probe submerged in reservoir
Telemetry field	`sensors.waterTemp`
Display precision	1 decimal place

4.2 Threshold Table

Range	Label	Status Code	Dashboard Color	Dashboard State	Alert Priority
< 16 °C	Too cold	warn	Red/Blue	Critical	Critical (see note)
16 – < 18 °C	Slightly cool	low	Blue (subtle)	Monitoring	Informational
18 – < 23 °C	Optimal	ok	Green	Stable	None
23 – < 25 °C	Slightly warm	low	Amber (subtle)	Monitoring	Informational
25 – < 27 °C	Warm	low	Amber	Monitoring	Informational
≥ 27 °C	Too warm — low O ₂ risk	warn	Red	Critical	Critical

4.3 Agronomist Rule Mapping

Rule ID	Condition	Severity	Priority	Recommended Action
waterTemp.too_warm	waterTemp ≥ 27 °C	critical	10	Move reservoir away from heat sources. Add cooled water at next refill.
waterTemp.warm	25 ≤ waterTemp < 27 °C	warning	11	Move reservoir away from heat sources. Add cooled water at next refill.
waterTemp.slightly_warm	23 ≤ waterTemp < 25 °C	info	12	Monitor root development. Reduce reservoir temperature if condition persists.

4.4 Alert Generation

Condition	Alert dedup key	Alert Severity
-----	-----	-----
waterTemp insight = 'warn'	wt-warn-{deviceId}	warning

4.5 Notes

- No dedicated agronomist rule exists for "too cold" (< 16 °C) water temperature in V1 from the `AGRONOMIST\_RULES` array. The `getSensorInsight()` function classifies it as 'warn' (status code) for display, but no dedicated V1 rule fires from the agronomist engine for this lower boundary. The notification alert (`wt-warn`) fires when insight status is 'warn', which includes the too-cold case.
- Water temperature lags behind air temperature in the simulation model. The thermal coupling uses a convergence rate of 5% per tick toward the target (air temperature – 0.8 °C offset). Firmware thermal modeling should incorporate a similar inertia characteristic.
- The critical upper threshold (27 °C) corresponds to the onset of oxygen depletion risk in the nutrient solution, which compromises root oxygenation and promotes pathogen development.

5. Air Temperature

5.1 Sensor

Attribute	Value
-----	-----
Parameter	Air temperature inside tower enclosure
Unit	Degrees Celsius (°C)
Sensor	Internal temperature sensor inside tower body
Telemetry field	`sensors.airTemp`
Display precision	1 decimal place

5.2 Threshold Table

Range	Label	Status Code	Dashboard Color	Dashboard State	Alert Priority
< 12 °C	Critical cold	warn	Red/Blue	Critical	Critical
12 – < 16 °C	Slightly cool	low	Blue (subtle)	Monitoring	Warning
16 – < 24 °C	Good for growth	ok	Green	Stable	None
24 – < 26 °C	Slightly warm	low	Amber (subtle)	Monitoring	Informational
26 – < 28 °C	Warm	low	Amber	Monitoring	Warning
≥ 28 °C	Heat stress risk	warn	Red	Critical	Warning

5.3 Agronomist Rule Mapping

Rule ID	Condition	Severity	Priority	Recommended Action
airTemp.too_cold	airTemp < 12 °C	critical	34	Raise temperature immediately. Provide supplemental heat or relocate.
airTemp.cool	12 ≤ airTemp < 16 °C	warning	33	Increase ambient temperature to bring into the optimal 18–24 °C range.
airTemp.slightly_warm	24 ≤ airTemp < 26 °C	info	32	Improve air circulation or reduce nearby heat sources.
airTemp.warm	26 ≤ airTemp < 28 °C	warning	31	Improve air circulation or reduce nearby heat sources to bring into 18–24 °C.
airTemp.too_hot	airTemp ≥ 28 °C	warning	30	Reduce temperature immediately. Increase ventilation or move tower.

5.4 Alert Generation

Condition	Alert dedup key	Alert Severity
airTemp.insight = 'warn'	at-warn-{deviceId}	warning
airTemp.insight = 'low'	at-low-{deviceId}	warning

5.5 Notes

- Note that `airTemp.too\_hot` (≥ 28 °C) has severity 'warning' in the agronomist rule engine, not 'critical'. However, the `getSensorInsight()` function classifies it as status 'warn' for display purposes. The health score impact is a –15 point deduction (warning penalty).
- The `airTemp.too\_cold` rule (< 12 °C) is classified 'critical' and imposes a –40 point health score deduction.
- Priority ordering: too_hot (30) is evaluated before cool (33). If both conditions were somehow simultaneously true, too_hot would appear first. In practice, only one boundary direction can be active at a time.

6. Relative Humidity

6.1 Sensor

Attribute	Value	
-----	-----	
Parameter	Relative humidity inside tower enclosure	
Unit	Percent (%)	
Sensor	Humidity sensor inside tower body	
Telemetry field	`sensors.humidity`	
Display precision	Integer (no decimal)	

6.2 Stage-Dependent Target Ranges

The humidity evaluation is growth-stage-dependent. The "too low" threshold is always `stage.min – 10`.

Growth Stage	RH Min	RH Max	RH Critical Low (min – 10)	
-----	-----	-----	-----	

Germination	80 %	96 %	70 %	
Seedling	70 %	96 %	60 %	
Vegetative	60 %	96 %	50 %	
Mature	50 %	96 %	40 %	

6.3 Threshold Table (Stage-Dependent)

Condition	Label	Status Code	Dashboard Color	Dashboard State	Alert Priority
-----	-----	-----	-----	-----	-----
RH < stage.min – 10	Very low internal RH	warn	Red	Critical	Critical
stage.min – 10 ≤ RH < stage.min	RH below stability target	low	Amber (subtle)	Monitoring	Warning
stage.min ≤ RH ≤ 96	Stable	ok	Green	Stable	None
96 < RH ≤ 99	High internal RH	low	Amber	Monitoring	Warning
RH > 99	Near-saturated internal RH	warn	Red	Attention	Warning

6.4 Threshold Table (Fallback — No Stage Configured)

Condition	Label	Status Code	Alert Priority
-----	-----	-----	-----
RH < 65	Very low internal RH	warn	Critical
65 ≤ RH < 80	RH below stability target	low	Warning
80 ≤ RH ≤ 99	Stable	ok	None
RH > 99	Near-saturated internal RH	warn	Warning

6.5 Agronomist Rule Mapping

Rule ID	Condition	Severity	Priority	Recommended Action
-----	-----	-----	-----	-----
humidity.too_low	RH < stage.min – 10 (or RH < 65 fallback)	critical	40	Check water reservoir level, irrigation cycle settings, and tower closure.
humidity.low	stage.min – 10 ≤ RH < stage.min	warning	41	Verify tower closure and irrigation frequency. Monitor ΔW trend.
humidity.high	stage.max < RH ≤ 99	warning	42	Monitor for condensation, stagnant air, unusual odours. Check air circulation.
humidity.too_high	RH > 99	warning	43	Monitor condensation on surfaces, stagnant air pockets, and unusual odours.

6.6 Alert Generation

Condition	Alert dedup key	Alert Severity
-----	-----	-----
humidity insight = 'warn'	rh-warn-{deviceId}	warning
humidity insight = 'low'	rh-low-{deviceId}	warning

6.7 Notes

- GL Towers internal microclimate RH values up to 96% are often compatible with system stability. Values above 96% but below 100% generate a warning-level alert but not a critical alert.
- The GL Towers tower enclosure creates a naturally elevated humidity microclimate due to recirculating irrigation and the evapotranspiration of growing plants. Target ranges reflect this enclosed environment, not ambient room conditions.
- For `humidity.too\_high` (RH > 99%), the physical maximum measurable is 100%. Values above 99% indicate sensor saturation or condensation on the sensor element.

7. Water Consumption

7.1 Sensor / Derived Parameter

Attribute	Value
Parameter	Estimated net daily plant water consumption
Unit	Milliliters per day (mL/day) — total across all plants
Source	Derived: plant count × stage consumption center value
Telemetry field	Derived from `sensors`, `savedPlants`, `savedIrrigation`

7.2 Net Consumption Ranges by Growth Stage

These are the observed net plant water uptake (ΔW) values per individual plant per day, used for reservoir depletion forecasting.

Growth Stage	Min (mL/day/plant)	Max (mL/day/plant)	Center (mL/day/plant)
Germination	1	3	2.0
Seedling	6	8	7.0
Vegetative	10	14	12.0
Mature	14	18	16.0

7.3 Net Daily Consumption Formula

...

$$\text{netDailyL} = (\text{centerValue_mL} \times \text{plantCount}) / 1000 \text{ [L/day]}$$

...

Fallback (irrigation not configured or stage undefined): `netDailyL = 0.015 L/day`

7.4 Stage Water Factors (for pump-based estimate, alternate method)

An alternate consumption estimate based on pump runtime (used internally for irrigation calculations):

Growth Stage	Stage Water Factor
Germination	0.5
Seedling	0.7
Vegetative	1.0
Mature	1.3

Formula: ``daily_L = pumpRuntimeMin × 0.10 L/min × stageFactor``

7.5 Dashboard Thresholds

No direct dashboard color/status is assigned to the consumption value itself. Consumption feeds the Refill Forecast (Section 8) which has defined thresholds.

7.6 Notes

- Net consumption (ΔW) is the water actually absorbed by plants, not the gross volume pumped through the system. Net consumption is always less than gross pump delivery.
- The ``GROWTH_STAGE_CONSUMPTION_RANGES`` values are GL Towers validated ranges under standard indoor conditions.

8. Water Level (Reservoir)

8.1 Sensor / Derived Parameter

Attribute	Value

Parameter	Reservoir water fill level	
Unit	Percent (%) and Liters (L)	
Telemetry field	`sensors.cwbeLevel` (percent), `dev.tankLiters` (liters)	
Physical limits	Min: 0 L, Max: 6.5 L	
Nominal capacity	6.0 L	
Pump flow rate	0.10 L/min	

8.2 CWBE Level Status Thresholds

Level Range	cwbeStatus Field	Display Label	Dashboard Color
-----	-----	-----	-----
≥ 30 %	ok	Optimal	Green
15 – < 30 %	low	Low warning	Amber
< 15 %	critical	Critical	Red

8.3 Refill Forecast Thresholds

Condition	Forecast Class	Display Text	Dashboard Color
-----	-----	-----	-----
Irrigation not configured	pending	"No irrigation data"	Neutral/grey
netDailyL ≤ 0	pending	"Refill forecast unavailable"	Neutral/grey
lvl ≤ 5 %	critical	"Low reservoir"	Red
daysRemaining < 1	critical	"Refill today"	Red
1 ≤ daysRemaining ≤ 3	soon	"Refill in ~N days"	Amber
daysRemaining > 3	ok	"Refill in ~N days"	Green

8.4 Alert Generation

Condition	Alert dedup key	Alert Severity
-----	-----	-----
cwbeLevel ≤ 5 %	reservoir-crit-{deviceId}	critical

daysRemaining < 1	refill-today-{deviceId}	critical	
1 ≤ daysRemaining ≤ 3	refill-soon-{deviceId}	warning	

8.5 Recommended Actions

Forecast Class	Recommended Action	
-----	-----	
critical	Refill the reservoir immediately before the next irrigation cycle.	
soon	Plan a refill soon to maintain stable irrigation.	
ok	No action required.	
pending	Configure irrigation to enable refill forecasting.	

8.6 Notes

- `criticalPct` = 5%: The critical alert fires when the level is at or below 5% of the physical maximum (6.5 L), equivalent to ≤ 0.325 L remaining.
- `soonDays` = 3: The "refill soon" alert fires when the estimated days-remaining is ≤ 3 calendar days.
- `cwbeCritPct` = 15% and `cwbeWarnPct` = 30% are used for the CWBE status field (`sensors.cwbeStatus`). These drive the color class of the reservoir hero block and badge.
- The 5% `criticalPct` alert threshold is separate from the 15% `cwbeCritPct` display threshold. The alert fires at 5%; the display turns red at 15%.

9. Pump Status

9.1 Parameter

Attribute	Value	
-----	-----	
Parameter	Water pump operational state	
Unit	Boolean (on / off)	

Telemetry field	`sensors.pumpActive`	
-----------------	----------------------	--

9.2 State Table

sensors.pumpActive	Status Label	Dashboard Color	Dashboard State	
-----	-----	-----	-----	
true	Running · manual	Green	Active	
false	Idle · on schedule	Grey	Inactive	

9.3 Alert Generation

No alerts are generated directly from pump state in V1. Pump state is displayed for informational purposes.

9.4 Operational States

State	Description	
-----	-----	
Running	Pump motor active. Water being circulated through the system.	
Idle	Pump motor stopped. Tower operating in dry interval between cycles.	

9.5 Notes

- Pump state is included in the telemetry snapshot. The dashboard reflects the hardware-reported state, not a computed state.
- Manual pump control via the dashboard dispatches a hardware command immediately, with the telemetry snapshot confirming the actual state on the next tick.

10. LED Status

10.1 Parameter

Attribute	Value	
-----	-----	
Parameter	LED grow light operational state	
Unit	Enum: 'active' / 'scheduled' / 'off'	
Source	Derived from savedLed + system clock	
Fields used	`savedLed.on`, `savedLed.off`, `savedLed.intensity`, `savedLed.preset`	

10.2 LED Status Determination Logic

```
...  
  
if intensity = 0 OR preset = 'off':  
    status = 'off'  
else:  
    onMins = onHour × 60 + onMinute  
    offMins = offHour × 60 + offMinute  
    nowMins = currentHour × 60 + currentMinute  
    if offMins ≤ onMins: status = 'scheduled' (overnight/24h cycle)  
    if nowMins ≥ onMins AND nowMins < offMins: status = 'active'  
    else: status = 'scheduled'  
...
```

10.3 State Table

Status	Condition	Dashboard Label	Dashboard Color	Badge Behavior	
-----	-----	-----	-----	-----	
active	Currently within on/off window	Active	Green	Animated dot	
scheduled	Outside on/off window	Scheduled	Amber	Static	
off	Intensity = 0 or preset = 'off'	Off	Grey	Static	

10.4 LED Presets

Preset Name	ON Time	OFF Time	Intensity	Daily Hours
Seedling	06:00	18:00	40 %	12 h
Standard	06:00	22:00	80 %	16 h
Bloom	05:00	23:00	100 %	18 h
Off	00:00	00:00	0 %	0 h
Custom	User-defined	User-defined	User-defined	Variable

10.5 Irrigation / LED Alignment Check

Condition	Alignment Status	Alert Severity
LED preset = 'off' or intensity = 0	ok	None
LED schedule is 24-hour ($\text{offMins} \leq \text{onMins}$)	ok	None
Night break = false (irrigation runs 24h)	warn	Warning
Irrigation window fully within LED hours	ok	None
Irrigation window extends outside LED hours	warn	Warning

Alert dedup key: `irr-align-{deviceId}`.

10.6 Notes

- The LED status is re-evaluated on every render cycle (it depends on the current wall-clock time, not on telemetry).
- A 24-hour LED cycle (where OFF time $\leq$ ON time) always returns 'scheduled' status in V1. True 24-hour operation detection is reserved for future implementation.

11. Connectivity / Telemetry Status

11.1 Parameter

Attribute	Value
Parameter	Communication link quality and telemetry freshness
Unit	Enum: 'online' / 'warning' / 'offline'
Source	Derived: `Date.now() - sensors.lastUpdate`
Fields used	`sensors.lastUpdate`, `metadata.connectionStatus`

11.2 Connection Status Threshold Table

Age of Last Update	connectionStatus	Alert Generated	Dashboard Color
$\leq 120,000$ ms (2 min)	online	None	Green dot
120,001 – 240,000 ms	warning	telemetry-stale (critical)	Amber
$> 240,000$ ms (4 min)	offline	telemetry-stale remains active	Red/Grey

11.3 Agronomist Offline Threshold

Age of Last Update	Effect
$\leq 300,000$ ms (5 min)	Agronomist analysis active
$> 300,000$ ms	Agronomist plant analysis suspended; offline warning displayed

11.4 Alert Generation

Condition	Alert dedup key	Alert Severity
lastUpdate age $> 120,000$ ms	telemetry-stale-{deviceId}	critical

11.5 Recommended Actions

Status	Recommended Action
-----	-----
warning	Check tower wireless connection. Ensure tower is powered and within range.
offline	Check tower connection. Verify power supply. Restart tower if necessary.

11.6 Notes

- When `metadata.overrideConnectionStatus` is set (for testing or manual override), the computed status is bypassed and the override value is used directly.
- Signal quality ('excellent', 'good', 'fair', 'poor') is reported in device metadata but does not generate alerts in V1.

12. Sensor Availability

12.1 Parameter

Attribute	Value
-----	-----
Parameter	Availability and validity of each sensor channel
Unit	Boolean per channel
Source	Presence and range-validity check on telemetry snapshot

12.2 Availability State Table

State Behavior	Condition	Display Behavior	Health Engine
-----	-----	-----	-----

Available evaluated	Value is numeric, within plausibility range	Normal display	Rules
Missing for this param	Value is null, NaN, or undefined	Display "--"	Rules suppressed
Invalid fault logged	Value outside plausibility range	Display "--"	Rules suppressed,
Pre-telemetry	lastUpdate = null (no snapshot ever received)	Display "--" for all sensors	All rules suppressed

12.3 Physical Plausibility Bounds

Sensor	Min Plausible	Max Plausible	
-----	-----	-----	
pH	0.0	14.0	
waterTemp	0 °C	50 °C	
airTemp	0 °C	50 °C	
humidity	0 %	100 %	
cwbeLevel	0 %	100 %	

12.4 Notes

- Sensor availability checks run on every telemetry snapshot, not on a separate timer.
- A "pre-telemetry" state persists until the first valid telemetry snapshot is received after boot. After first snapshot, individual sensor channels can become "missing" or "invalid" independently.

13. Health Score Bands

13.1 Score Range Mapping

Score Range	Status Label	Dashboard Badge Color	Agronomist Summary

-----	-----	-----	-----
85 – 100	Excellent	Green	"The system appears stable. All monitored parameters are within the expected range."
65 – 84	Good	Green	"The system is generally operating well. Minor observations may require monitoring."
40 – 64	Attention	Amber	"The system requires attention. One or more parameters are outside the preferred operating range."
0 – 39	Critical	Red	"Immediate action is recommended. One or more critical conditions may affect plant stability."
—	No Data	Grey	No telemetry received yet.

13.2 Final Status Override Rule

The health status is always the ****worse**** of the score-derived status and the rule-derived status:

Any Critical Rule Fired	Any Warning Rule Fired	Any Info Rule Fired	Score Status	Final Status
-----	-----	-----	-----	-----
Yes	—	—	Any	Critical
No	Yes	—	≥ 65	Attention
No	No	Yes	≥ 85	Good
No	No	No	≥ 85	Excellent
No	No	No	65–84	Good
No	Yes	—	40–64	Attention
No	No	Yes	40–64	Attention

14. Alert Severity Penalties

14.1 Penalty Table

Severity	Penalty per Fired Rule	Notes	
----------	------------------------	-------	--

-----	-----	-----
critical	- 40 points	One critical rule → score ≤ 60
warning	- 15 points	One warning rule → score ≤ 85
info	- 5 points	One info rule → score ≤ 95

14.2 Cumulative Penalty Examples

Fired Rules	Total Deduction	Score	Status	
-----	-----	-----	-----	
No rules	0	100	Excellent	
1 info rule	- 5	95	Excellent	
2 info rules	- 10	90	Excellent	
1 warning rule	- 15	85	Excellent (boundary)	
1 warning + 1 info	- 20	80	Good	
2 warning rules	- 30	70	Good	
1 critical rule (from rules)	- 40	60	Attention (from score) → overridden to Critical	
2 critical rules	- 80	20	Critical	
1 critical + 2 warning + 1 info	- 75	25	Critical	

14.3 Notes

- The score floor is 0. A score cannot go below 0 regardless of how many rules fire.
- The rule-derived status overrides the score-derived status when it is worse. This means a single critical rule always results in status 'Critical', even if the score is still in the 'Attention' range.

15. Telemetry Timing Thresholds

15.1 Complete Timing Reference

Parameter	Value	Unit	Purpose	
-----	-----	-----	-----	
Telemetry tick — Slow	30,000	ms	Sensor update interval (slow speed)	
Telemetry tick — Normal	15,000	ms	Sensor update interval (default)	
Telemetry tick — Fast	5,000	ms	Sensor update interval (fast speed)	
History sample interval	10,000	ms	Minimum gap between recorded history samples	
History max samples	480	—	Rolling buffer maximum size	
Persist interval	Every 4 ticks	—	Device state and history written to storage	
Notification throttle	8,000	ms	Minimum interval between notification generation runs	
Event notification throttle	10,000	ms	Window for suppressing duplicate event alerts	
Stale data threshold	120,000	ms	Triggers CommunicationWarning state and stale alert	
Offline threshold	240,000	ms	Triggers CommunicationOffline state	
Agronomist offline threshold	300,000	ms	Suspends plant analysis	
Max notifications stored	50	—	Alert log capacity	

16. Confidence Bands

16.1 Complete Confidence Reference

The agronomist confidence score describes the reliability of the analytical assessment based on the depth of available telemetry history.

Minimum Samples	Coverage Hours Required	Confidence Score	Label	
Meaning				
-----	-----	-----	-----	-----
< 3 samples history	—	0.00	None	Snapshot only; insufficient

≥ 3 samples reliable	< 1 hour	0.30	Live	Recent data only; trends not yet
≥ 3 samples available	1 – < 6 hours	0.55	Partial history	Short-term trends
≥ 3 samples available	6 – < 24 hours	0.80	Limited history	Medium-term trends
≥ 3 samples available	≥ 24 hours	1.00	Extended history	Full trend analysis

16.2 Notes

- Coverage is computed as: `(samples[-1].t - samples[0].t) / 3,600,000` (hours).
- At the default 15 s telemetry interval with 10 s minimum sample gap, approximately 240 samples cover 2 hours of operation (which yields 'Limited history' confidence). Reaching 'Extended history' confidence requires sustained operation for 24 hours.
- The confidence score is not used in health scoring directly. It is displayed to the operator as context for the reliability of trend and recommendation outputs.

17. Trend Detection Thresholds

17.1 Algorithm Parameters

Parameter	Value	
-----	-----	
Trend window	Last 10 samples	
Minimum samples required	3	

17.2 Trend Delta Thresholds

A trend is reported as 'rising' or 'falling' only when the mean of the second half of the window differs from the mean of the first half by at least the delta threshold.

Parameter	Delta Threshold	Unit	Trend Labels
-----	-----	-----	-----
pH	0.20	pH	rising / falling / stable
Water Temp	0.50	°C	rising / falling / stable
Air Temp	0.50	°C	rising / falling / stable
Humidity	3.00	%	rising / falling / stable
CWBE Level	3.00	%	rising / falling / stable

17.3 Trend Outcome Table

Delta	Trend Result
-----	-----
delta > threshold	rising
delta < -threshold	falling
-threshold ≤ delta ≤ threshold	stable
< 3 samples available	unknown

17.4 Notes

- Trend analysis is computed only within `getAgronomistReport()`. It does not affect health scoring, alert generation, or any operational decision in V1.
- Trend results are displayed as qualitative text in the Agronomist panel.

18. Irrigation Thresholds

18.1 Irrigation Matrix

Complete GL Towers validated preset table. Values: ON (min) / OFF (min). Baseline: Italy indoor, March/April, 18 °C, 0.10 L/min pump.

Stage	Soil	Coco Coir	Hydroponic	Clay Pebbles	Rockwool
Germination	1 / 360	1 / 300	1 / 240	1 / 210	1 / 300
Seedling	1 / 180	1 / 150	1 / 120	1 / 105	1 / 180
Vegetative	1.5 / 150	1.5 / 120	1.5 / 90	1.5 / 75	1.5 / 150
Mature	2 / 90	2 / 75	2 / 60	2 / 50	2 / 90

18.2 Cycle Pause Bounds (Applied by Recommended Cycle Calculator)

Bound	Value
Minimum	15 min
Maximum	360 min

18.3 Temperature Category Modifier for Off-Cycle

Category	Air Temp Range	Multiplier Applied to OFF Time
Cool	< 18 °C	× 1.15
Normal	18 – 23 °C	× 1.00
Warm	23 – 28 °C	× 0.82
Hot	> 28 °C	× 0.67

18.4 Crop Demand Divisor for Off-Cycle

Crop Category	Divisor	Effect
Leafy greens	1.00	No modification
Basil	0.95	Slightly shorter pause (more frequent)
Parsley	0.80	Shorter pause
Herbs	0.73	Shorter pause

Microgreens	0.68	Shortest pause (most frequent)	
Fruiting	1.20	Longest pause (highest water demand per cycle)	

18.5 Substrate-Specific Cycle Warnings

Substrate	Condition	Warning Threshold	
-----	-----	-----	
Soil	Pause too short	cyclePause < 30 min	
Hydroponic	Pause too long	cyclePause > 60 min	
Clay	Pause too long	cyclePause > 45 min	

18.6 Night Break Window

When `nightBreak = true`, irrigation cycles are constrained to the configured window (windowStart–windowEnd). Cycles that would fall outside the window are suppressed.

Default window: 06:00 – 22:00.

18.7 Cycles Per Day Formula

...

period = cycleOn + cyclePause [minutes]

if nightBreak AND windowStart AND windowEnd:

 windowMins = irrWindowMins(windowStart, windowEnd)

 cyclesPerDay = max(0, floor(windowMins / period))

else:

 cyclesPerDay = floor(1440 / period)

...

18.8 Runtime Per Day

...

$\text{runtimePerDay} = \text{cyclesPerDay} \times \text{cycleOn} \text{ [minutes/day]}$

...

19. Crop-Specific Adjustments

19.1 Crop Classification for Irrigation

Crop ID Contains	Crop Key
-----	-----
'microgreen'	microgreens
'basil'	basil
'parsley'	parsley
'tomato', 'pepper', 'chili', 'cucumber', 'zucchini', 'eggplant', 'strawberry'	fruiting
'rosemary', 'thyme', 'oregano', 'sage', 'lavender'	herbs
(default)	leafy

Classification is case-insensitive substring matching on the plant ID.

20. LED Preset Reference

20.1 Complete Preset Values

Preset	ON Time	OFF Time	Intensity	Daily Photoperiod
-----	-----	-----	-----	-----
Seedling	06:00	18:00	40 %	12 h

Standard	06:00	22:00	80 %	16 h	
Bloom	05:00	23:00	100 %	18 h	
Off	00:00	00:00	0 %	0 h	

20.2 Preset Detection

A saved LED schedule is classified as a named preset only if all three values (ON, OFF, intensity) exactly match a preset entry. If any value differs, the schedule is labeled 'Custom'.

20.3 Notes

- Intensity is expressed as an integer percentage (0–100).
- ON and OFF times are stored as HH:MM strings (24-hour format).
- The daily photoperiod calculation: $\text{hours} = \max(0, (\text{offMins} - \text{onMins}) / 60)$. This does not handle overnight schedules spanning midnight in V1.

21. CWBE — Reserved for Future Implementation

21.1 Status

CWBE (Crop Water Balance Efficiency) is an advanced scientific module currently under validation.

****CWBE is intentionally excluded from Dashboard V1.****

The dashboard architecture has been designed to allow future integration of CWBE without requiring structural redesign.

21.2 Reserved Threshold Positions

The following threshold zones are reserved for CWBE integration and will be populated when the CWBE module is validated:

Threshold Area	Status
-----	-----
CWBE efficiency score ranges	Reserved for future implementation
CWBE alert priority ranges	Reserved for future implementation (priority 60–69 in rule engine)
CWBE health score penalty values	Reserved for future implementation
CWBE trend thresholds	Reserved for future implementation
CWBE correlation thresholds	Reserved for future implementation

21.3 Current Placeholder

The ``sensors.cwbeLevel`` field currently carries the reservoir fill percentage (0–100%). This field is named for future CWBE integration. The telemetry history records this field in every sample, providing continuity when the CWBE efficiency metric replaces or supplements it.

The ``sensors.cwbeStatus`` field (ok / low / critical) is currently derived from ``cwbeLevel`` against the 15% and 30% thresholds. These thresholds will be revised when CWBE is integrated.

End of Document — GL Towers Dashboard Operational Threshold Specification v1.0